

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

III Semester of M.Sc. Physic Examination November 2019

PS907 Thin Film Technology

Date: 18-11-2019 Day: Monday Time: 1.30 PM To 1:50 PM Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

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- Q-I Choose the correct answer for the following questions. 10**
1. PLD method uses to vaporize the target material for the deposition of thin film. 1
(a) electron beam (b) laser beam
(c) thermal energy (d) photon beam
 2. In the case of RF sputtering process, the size of the target is chosen to be then the substrate in order to minimize the substrate sputtering during deposition 1
(a) bigger (b) equal
(c) smaller (d) comparable
 3. Select the dominant gas flow regime, in the chamber, during high vacuum condition 1
(a) Molecular flow (b) Intermediate flow
(c) Viscous flow (d) all of above
 4. The pump which is useful to create the vacuum $\sim 10^{-10}$ torr or better in the chamber. 1
(a) rotary pump (b) turbo pump
(c) diffusion pump (d) ion pump

5. How a uniform film with good crystal structure is attained in cathode sputtering process? 1
(a) By hitting high energy particle directly on the substrate
(b) Allowing Less time for the particles to deposit on the substrate
(c) High energy particle diffuse through low pressure gas and deposits on the substrate
(d) Heavy inert gas is used for film deposition on the substrate

6. Fine grained, smooth and continuous thin film growth requires 1
(a) low supersaturation (b) high deposition rate
(c) High supersaturation (d) low substrate temperature

7. A technique utilize an atomizer for creating fine droplets of the precursor solution during thin film deposition. 1
(a) sol gel method (b) dip coating
(c) spray pyrolysis (d) electroplating

8. Usually the vaporization of the target material has been carried out under high vacuum in order to 1
(a) minimize the impurity level
(b) reduce the scattering of vaporized species
(c) to increase the mean free path of vaporized species
(d) above all

9. reactions, in the CVD growth method, are responsible for the inclusion of lots of defect in grown film. 1
(a) Homogeneous (b) heterogeneous
(c) both (a) and (b) (d) none of this

10. In which of the following process, a non-thermal physical energy is involved in the vaporization of the target material through bombardment of gaseous ions, 1
(a) MBE (b) PLD
(c) sputtering (d) e-beam deposition

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Date: 18-11-2019 Day: Monday Time: 1.50 PM To 3:30 PM Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION – I

Q – II Answer the following questions as directed. 10

1. Explain importance of substrate cleaning prior to film deposition. 2
2. Discuss the different types of gas flow during vacuum creation. 2
3. Define following terms 2
 - (i) Sputtering
 - (ii) Pumping speed
4. List the advantages and applications of the chemical vapor deposition method. 2
5. Write short note on working of rotary vane pump 2

SECTION – II

Q-III Answer the following questions as directed (Any three) 15

1. Find the critical radius for homogeneous nucleation, at room temperature, having atomic volume $2 \times 10^{-23} \text{ cm}^3/\text{at}$, $\gamma = 1000 \text{ ergs/cm}^2$, $P_v = 10^{-3} \text{ torr}$ and $P_s = 10^{-10} \text{ torr}$. (Boltzman constant $= 1.380649 \times 10^{-16} \text{ erg} \cdot \text{K}^{-1}$) 5
2. Discuss in detail the dip coating method, its advantages and disadvantages. 5
3. Discuss the four stages of thin film growth. 5
4. Discuss in detail the sputtering process. 5